

Thermodynamic Waste, Health Capital, and Long-Run Growth

A Thermo-Ecological Augmentation of the Solow Model

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Abstract

This paper develops a thermo-ecological augmentation of the Solow growth model that treats production as a physical throughput process generating both useful output and unavoidable thermodynamic waste. We introduce three coupled state variables absent from the standard framework: (i) **physical thermodynamic output** (TO), the energy waste, emissions, and material inefficiency created as a byproduct of production, valued at shadow prices derived from the social cost of carbon and analogous pollutant valuations; (ii) **health capital** (H), a stock variable linking pollution exposure to effective labor supply, cognitive performance, and total factor productivity; and (iii) **natural capital** (R), resource stocks subject to depletion and regeneration dynamics.

We derive the modified Solow steady state in closed form and show that it depends on a **thermo-ecological multiplier** $\Omega = \phi(\text{TO}/Y) \cdot \zeta(H) \cdot \theta(H)$ that is strictly less than unity when thermodynamic waste and health damage are positive. The steady-state capital intensity and output per worker are reduced by a factor that depends on pollution intensity, health capital, and the elasticity of damage. We characterize the **thermodynamic growth trap**: economies that allow waste intensity to rise converge to permanently lower balanced growth paths. We define **Thermo-GDP** (a damage-adjusted output measure) and connect it to existing green accounting frameworks (SEEA, Adjusted Net Savings, Inclusive Wealth). Extensions to Ramsey optimal growth, Hotelling resource extraction, and DSGE environmental shocks are previewed for companion papers.

Keywords: economic growth, ecological economics, thermodynamics, health capital, environmental macroeconomics, productivity

JEL Codes: O44, Q56, E13, Q57

1. Introduction

Modern economies convert low-entropy resources into high-entropy products, services, and waste at unprecedented scale. The Second Law of Thermodynamics guarantees that every production process generates some waste heat and disorder. This does not impose a hard ceiling on growth, efficiency can improve enormously before approaching theoretical limits, and knowledge-intensive and service-based production have very different throughput profiles than heavy industry. What it does impose is an *irreducible physical cost* that varies with technology, energy mix, and economic structure, and that standard macroeconomic models systematically under-account for. ^[1] Canonical growth models from Solow (1956) to modern DSGE treat the economy as a closed circular flow of income, with environmental impacts relegated to externality corrections applied after the fact.

This omission has become untenable. The OECD projects that outdoor air pollution alone could reduce global GDP by approximately 1% by 2060 through labor productivity losses and healthcare expenditures. ^[2] The IPCC AR6 warns that unmitigated warming could reduce global economic output by 10–23% by 2100. ^[3] Empirical work in thermoeconomics has demonstrated that useful exergy explains a far larger share of historical growth than conventional capital–labor models suggest, substantially narrowing the Solow residual. ^[4,5]

The goal of this paper is to **augment the Solow growth model** with state variables that make the costs of thermodynamic waste, health degradation, and resource depletion visible within the standard framework. The augmentation is designed to be conservative: when environmental costs are zero, the model collapses exactly to the textbook Solow result. When those costs are positive, the model generates a lower steady-state growth path, providing a formal basis for the intuition that dirty growth today cannibalizes future output.

We focus on the Solow model because it is the canonical framework through which economists think about long-run growth, and because its simplicity makes the augmentation transparent. Extensions to the Ramsey–Cass–Koopmans optimal growth framework (where the discount rate becomes endogenous to environmental conditions), the Hotelling exhaustible resource model (where the optimal extraction path accounts for thermodynamic damage costs), and the New Keynesian DSGE framework (where environmental degradation generates a new class of supply shocks) are previewed in Section 7 and developed in companion papers.

1.1 What This Framework Does Not Claim

To prevent misreading, we state explicitly what this paper does *not* argue:

- It does not imply hard thermodynamic limits to near-term economic growth. Efficiency improvements, compositional shifts, and innovation can reduce TO intensity substantially.
- It does not claim that pollution is the dominant driver of the productivity slowdown. We propose it as a testable contributing mechanism, not a monocausal explanation.
- It does not replace standard macroeconomics. It augments the Solow model with additional state variables; when those variables are at baseline, the standard model obtains exactly.
- It does not reject technological progress or decoupling. Clean growth (rising output with falling TO intensity) is the policy objective the framework identifies.
- It does not claim GDP is useless. It proposes Thermo-GDP as a complementary dashboard metric, not a replacement.

1.2 Relation to Existing Literature

This framework builds on Brock and Taylor’s Green Solow Model, ^[15] Nordhaus’s DICE integrated assessment model, ^[18] and the broader IAM tradition. It differs from these predecessors in four specific ways. First, health capital enters as an endogenous state variable affecting *both* labor quality and TFP simultaneously, creating a double feedback channel absent from standard IAMs. Second, thermodynamic waste is modeled as a production-side throughput variable ($\tau = \text{TO}/Y$) rather than purely as downstream damage. The sustainability multiplier $\phi(\tau)$ operates on the production function directly, not as a post-hoc damage subtraction. Third, the Ω amplification mechanism—environmental degradation reducing steady-state output by *more* than the direct penalty because it also depresses equilibrium capital accumulation—is a structural result not present in Green Solow or DICE. Fourth, Thermo-GDP is internally linked to the same shadow-price structure as the production model, ensuring consistency between the growth framework and the accounting framework. The contribution is not that any single element is entirely new, but that their integration into a single analytically tractable model produces emergent results (the amplification effect, the growth trap) that do not appear in the separate literatures.

2. Theoretical Architecture

2.1 The Economy as a Dissipative Structure

Following Georgescu-Roegen (1971), we conceptualize the economy as an open dissipative system that maintains internal complexity by importing low-entropy resources and exporting high-entropy waste.^[4] This implies that production is *throughput* (requiring continuous material and energy flows), waste generation is *endogenous* (a physical law, not a market failure), and sustainability requires that the biosphere's finite *sink capacity* not be exceeded.^[6]

2.2 The Production Function

We adopt a four-factor production function that nests the standard Cobb–Douglas as a special case. Following Ayres and Warr (2005, 2009), who showed that including useful exergy as a factor of production substantially resolves the Solow residual,^[4,5] and Santos et al. (2018), who found exergy output elasticities of 0.29–0.46 across multiple countries,^[7] we write:

$$Y_t = A_t \cdot K_t^\alpha \cdot L_t^\beta \cdot U_t^u \quad \text{where } \alpha + \beta + u = 1 \quad (1)$$

where K_t is physical capital, L_t is labor, U_t is useful exergy (primary exergy multiplied by conversion efficiency f_t), and A_t is residual TFP. In the core model (Section 3), we work with the standard two-factor reduction $Y = AK^\alpha(AL)^{1-\alpha}$ for analytical tractability, treating exergy as embedded in TFP. The four-factor version is used for simulation and empirical calibration.

2.3 Physical Thermodynamic Output

We define physical thermodynamic output as the aggregate material and energy waste generated by production. Crucially, TO is measured in **physical units** (tonnes, joules, cubic meters)—it is a vector of waste flows, not a damage measure:

$$TO_t = (E_t, P_t, M_t) \quad (2)$$

where:

- E_t = energy waste: lost heat, flaring, transmission losses (joules)
- P_t = pollution emissions: CO₂e, PM_{2.5}, NO_x, SO₂, heavy metals, wastewater (tonnes)
- M_t = material waste: landfill volumes, tailings, single-use plastics (tonnes)

Note what is *not* in TO: health damage, ecosystem damage, and welfare losses. These are **downstream consequences** of physical TO, not components of it. This separation is essential to avoid double-counting. The causal chain runs:

Production → Physical TO (waste flows) → Shadow-priced damage → Health effects →
Feedback to production

We define **TO intensity** as a dimensionless scalar index of waste-per-unit-output:

$$\tau_t \equiv \|TO_t\| / (p_0 \cdot Y_t) \quad (3)$$

where $\|TO_t\|$ is a weighted norm that aggregates across waste types using damage-equivalence factors (e.g., global warming potentials convert all greenhouse gases to CO_{2e} tonnes), and p_0 is a fixed normalization constant with units of [currency per tonne-equivalent] chosen so that τ is dimensionless and bounded in $[0, 1]$. This is the same dimensional bridge that applied climate economists already use when expressing damages as “% of GDP”: physical flows are implicitly converted to a common monetary scale via a reference price. We make this normalization explicit. TO intensity is a *production-side* measure of how dirty the economy’s technology is. It is distinct from, and not inflated by, cleanup spending.

2.4 Damage Valuation

Physical TO causes economic damage through environmental degradation and health effects. We value this damage using shadow prices derived from the established social-cost literature:

$$D_t = \lambda' \cdot TO_t = \lambda_e E_t + \lambda_p P_t + \lambda_m M_t \quad (4)$$

where λ_n is the marginal social damage per unit of waste type n . For greenhouse gases, the EPA’s 2023 central estimate is ~\$190/tCO₂ at a 2% discount rate, ^[9] with Rennert et al. (2022) arriving at ~\$185/tCO₂. ^[10] Analogous values exist for methane (~\$1,600/tCH₄), nitrous oxide (~\$5,400/tN₂O), and PM_{2.5} (valued through mortality and morbidity costs). ^[9]

The total damage D_t has units of dollars per period. It represents the present value of all downstream harm: climate damage, ecosystem degradation, *and* health effects caused by one period’s physical waste flows. Health damage is a **subset** of D_t , not an additive component alongside it. This layered structure (physical flows, then prices, then downstream channels) eliminates the double-counting risk that arises when health damage appears both as a TO component and as a separate variable.

2.5 Health Capital

Health capital H_t is a state variable representing the productive capacity of the population’s health. It is driven down by pollution exposure (a downstream consequence of physical TO) and driven up by preventive health investment and baseline medical progress. The empirical foundations are strong: Graff Zivin and Neidell

(2012) found that ozone exposure significantly reduced agricultural worker productivity, [11] while European firm-level studies estimate that a 1 $\mu\text{g}/\text{m}^3$ PM_{2.5} increase reduces labor productivity by approximately 0.55%. [12]

Per-Capita Exposure

We define an exposure index that translates aggregate pollution into a per-person dose proxy:

$$X_t = \kappa \cdot P_t / \text{Pop}_t \quad (5)$$

where P_t is the pollution component of physical TO (Equation 2), κ is a dose–response scaling parameter calibrated to epidemiological data, and Pop_t is population. Note that X_t depends on *physical pollution flows*, not on monetized damage—maintaining the clean causal chain.

Health Stock Dynamics

The law of motion for health capital is:

$$\dot{H}_t = \eta_0 + \eta_1 h_t - \delta_h H_t - \chi X_t \quad (6)$$

where:

- η_0 = baseline health improvement from medical/sanitation progress
- $h_t = \text{HE}_t^{\text{prev}} / Y_t$ = preventive health spending as a share of output
- δ_h = natural health depreciation (aging, chronic disease)
- χX_t = health damage from environmental exposure, $\chi > 0$

In steady state, $H^* = (\eta_0 + \eta_1 h^* - \chi X^*) / \delta_h$. Higher pollution exposure (X) reduces steady-state health capital; higher preventive spending (h) raises it.

Health-Adjusted Inputs

Health capital enters the production function through two channels:

$$L_t^{\text{eff}} = L_t \cdot \theta(H_t) \quad \text{and} \quad A_t^{\text{eff}} = A_t \cdot \zeta(H_t) \quad (7)$$

where $\theta(H) = (H/H)^b$ captures labor quality (absenteeism, physical capacity, work hours) and $\zeta(H) = (H/H)^m$ captures the TFP channel (cognition, error rates, learning capacity), with $b, m > 0$ and H = baseline health. When $H = H$, both functions equal unity and the standard production function obtains. This creates the **feedback loop** at the heart of the model: production $\rightarrow$ pollution $\rightarrow$ exposure $\rightarrow$ health decline $\rightarrow$ reduced effective inputs $\rightarrow$ lower output.

The TFP channel $\zeta(H)$ is worth flagging: if pollution has been eroding cognitive performance in advanced economies, then measured TFP could be declining even as technological knowledge accumulates. This offers a testable contributing mechanism (not a complete explanation) for the observed productivity slowdown. A companion paper develops this hypothesis with firm-level data.

2.6 Sustainability Metrics

Following the critique that a single index can conflate production efficiency with remediation effort, we define **two separate** sustainability indicators:

TO Intensity (production-side cleanliness):

$$\tau_t = \|TO_t\| / (p_0 \cdot Y_t) \quad (3, \text{restated})$$

This measures how much physical waste is generated per unit of output. It depends on technology, energy mix, and process efficiency. It is *not* affected by cleanup spending. Lower is better.

Net Ecological Position (balance-sheet sustainability):

$$NEP_t = G(R_t) + C_t^{\text{restore}} - D_t \quad (8)$$

where $G(R_t)$ is natural regeneration, C_t^{restore} is restoration/abatement investment, and D_t is monetized damage (Equation 4). $NEP > 0$ means the economy is building natural capital on net; $NEP < 0$ means it is drawing down. This is analogous to the World Bank's Adjusted Net Savings but grounded in our damage-pricing framework. ^[21]

The Sustainability Multiplier. For the core Solow model, we define the sustainability multiplier as a function of TO intensity:

$$\phi(\tau_t) = (1 - \tau_t)\psi \quad \text{where } \psi > 0 \quad (9)$$

Axiomatic justification. We require ϕ to satisfy four properties: (A1) $\phi(0) = 1$ —zero waste imposes no penalty; (A2) ϕ is strictly decreasing in τ —more waste means less effective output; (A3) ϕ is convex—marginal damage is increasing; and (A4) as τ approaches its upper normalization bound, effective productive efficiency approaches zero. The family $(1 - \tau)\psi$ is the simplest function satisfying all four axioms. The parameter ψ controls the convexity of the penalty: $\psi < 1$ yields a gentle curve (small increases in waste have modest effects), $\psi = 1$ is linear, and $\psi > 1$ generates a steep penalty for dirty economies.

We emphasize that the qualitative results of the model, the amplified steady-state reduction and the signs of all comparative statics, hold for *any* ϕ that is decreasing, convex, and satisfies $\phi(0) = 1$. The specific power form is chosen for tractability. The

parameter ψ is, in principle, empirically estimable using cross-country regressions of TFP on TO intensity; a calibration strategy is outlined as a direction for future work. Note that ϕ depends only on TO intensity, not on restoration spending, resolving the perverse-incentive problem identified in the earlier formulation.

2.7 Natural Capital Dynamics

The stock of a renewable resource R_t evolves as:

$$dR_t/dt = G(R_t) - C_t \quad (10)$$

where $G(R) = rR(1 - R/K)$ is logistic regeneration with carrying capacity K and intrinsic growth rate r , and C_t is extraction. Sustainable extraction requires $C_t \leq G(R_t)$ in the long run. For non-renewable resources ($G \approx 0$), the Hotelling rule governs the optimal extraction path.^[14] Our framework modifies the Hotelling condition by adding the marginal thermodynamic damage cost of extraction to the user cost. However, the correct sign and functional form of this modification require a careful Hamiltonian derivation that is beyond the scope of the present paper. **Figure 1** summarizes the model architecture and the causal pathways linking production, thermodynamic output, health capital, and long-run growth.

3. The Core Model: Thermo-Ecological Solow

We now assemble the components into a single, formally specified growth model.

3.1 Setup

We work with the standard Solow assumptions: exogenous savings rate s , capital depreciation rate δ , labor force growth rate n , and Harrod-neutral technical progress at rate g . The production function is modified by the thermo-ecological multiplier:

$$Y_t = \Omega_t \cdot F(K_t, A_t L_t) = \Omega_t \cdot K_t^\alpha (A_t L_t)^{1-\alpha} \quad (11)$$

where $\Omega_t \equiv \phi(\tau_t) \cdot \zeta(H_t) \cdot [\theta(H_t)]^{1-\alpha}$ is the thermo-ecological multiplier combining the sustainability penalty and both health channels. Note:

- $\phi(\tau_t) = (1 - \tau_t)\psi$ — sustainability multiplier (Eq. 9)
- $\zeta(H_t) = (H_t / \bar{H})^m$ — health–TFP channel
- $\theta(H_t) = (H_t / \bar{H})^b$ — health–labor quality channel (enters with exponent $1-\alpha$ from the production function)

Capital accumulates in the standard way:

$$\dot{K}_t = sY_t - \delta K_t \quad (12)$$

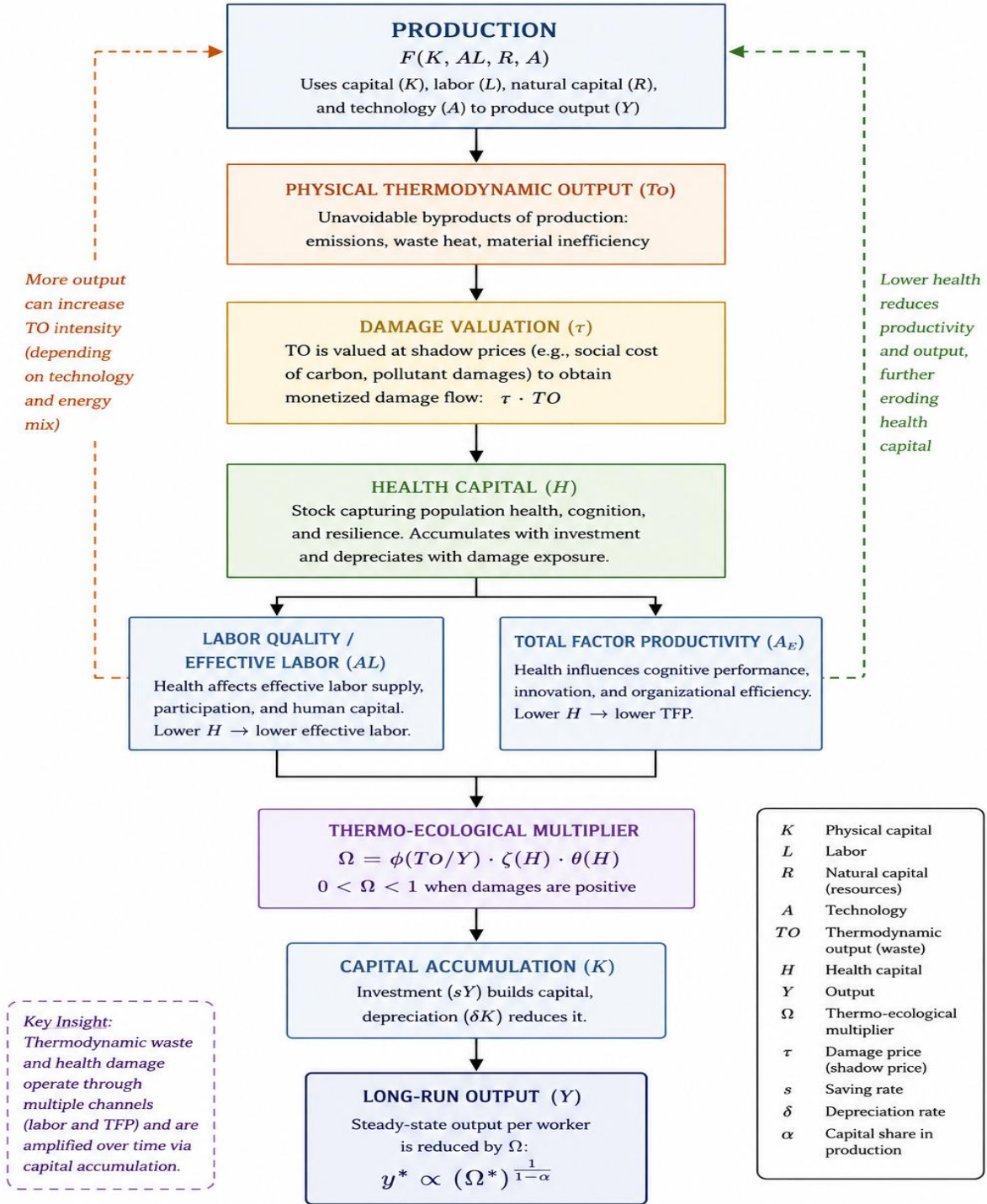


Figure 1. Thermo-Ecological Architecture of the Model. Production generates physical thermodynamic output (TO), which creates downstream environmental and health damages. Health capital affects both labor quality and total factor productivity, feeding through the thermo-ecological multiplier Ω into capital accumulation and long-run output.

3.2 Intensive Form

Define capital per effective worker: $k_t \equiv K_t / (A_t L_t)$. Output per effective worker is:

$$y_t = \Omega_t \cdot k_t^\alpha \quad (13)$$

The fundamental Solow equation in intensive form becomes:

$$\dot{k}_t = s \cdot \Omega_t \cdot k_t^\alpha - (n + g + \delta) \cdot k_t \quad (14)$$

This is identical to the standard Solow equation except for the Ω_t term multiplying the production function. When $\Omega = 1$ (no environmental costs), Equation (14) reduces to the textbook version.

3.3 Steady State

At the steady state $\dot{k} = 0$:

$$s \cdot \Omega^* \cdot (k^*)^\alpha = (n + g + \delta) \cdot k^* \quad (15)$$

Solving for k^* :

$$k^* = [s \cdot \Omega^* / (n + g + \delta)]^{1/(1-\alpha)} \quad (16)$$

Steady-state output per effective worker:

$$y^* = \Omega^* \cdot [s \cdot \Omega^* / (n + g + \delta)]^{\alpha/(1-\alpha)} \quad (17)$$

which simplifies to:

$$y^* = (\Omega^*)^{1/(1-\alpha)} \cdot [s / (n + g + \delta)]^{\alpha/(1-\alpha)} \quad (18)$$

This is the central result. Equation (18) is the standard Solow steady-state output *multiplied by* $(\Omega^*)^{1/(1-\alpha)}$. Since $0 < \Omega \leq 1$ and $1/(1-\alpha) > 1$, the thermo-ecological multiplier has an **amplified** effect on steady-state output: a 10% reduction in Ω reduces y^* by more than 10% because it also depresses equilibrium capital accumulation.

3.4 Decomposing the Multiplier

Expanding Ω^* :

$$\Omega^* = (1 - \tau^*)\psi \cdot (H^*/\bar{H})^m \cdot (H^*/\bar{H})^{b(1-\alpha)} \quad (19)$$

This reveals three distinct drags on the steady state:

- **Waste drag:** $(1 - \tau^*)\psi$. Higher TO intensity (dirtier technology) lowers potential output.
- **Cognitive drag:** $(H^*/\bar{H})^m$. Pollution-induced health deterioration impairs TFP through reduced cognition, increased errors, slower learning.

- **Labor drag:** $(H^*/H)^{b(1-\alpha)}$. Health deterioration reduces effective labor supply through absenteeism, shorter work hours, reduced physical capacity.

In the standard Solow model ($H = \bar{H}$, $\tau = 0$), all three terms equal unity, $\Omega = 1$, and the standard steady state obtains. This confirms that the augmented model nests the original as a special case.

3.5 Comparative Statics

Differentiating Equation (16) with respect to the key environmental parameters:

$$\partial k^*/\partial \tau < 0 \text{ — dirtier production} \rightarrow \text{lower capital intensity} \quad (20a)$$

$$\partial k^*/\partial H > 0 \text{ — healthier population} \rightarrow \text{higher capital intensity} \quad (20b)$$

$$\partial k^*/\partial \psi < 0 \text{ (for } \tau > 0) \text{ — steeper penalty} \rightarrow \text{lower steady state} \quad (20c)$$

These signs are unambiguous and match economic intuition. A complete derivation is provided in Appendix B.

3.6 The Thermodynamic Growth Trap

The growth trap arises when Ω_t is *declining over time*; i.e., when TO intensity is rising and/or health capital is falling. In this case, the steady state itself is moving downward: the economy is converging toward a target that is retreating. Short-run GDP growth remains positive (the economy is still accumulating capital), but the long-run balanced growth path is lower than it would be under stable environmental conditions.

Critically, the trap is **invisible in conventional GDP**. An economy in the trap shows positive output growth, normal unemployment, and no obvious distress while its sustainable productive capacity is silently eroding. Only a damage-adjusted measure like Thermo-GDP (Section 4) reveals the deterioration. This is perhaps the framework's most policy-relevant finding.

This mechanism generates a testable contributing hypothesis for **secular stagnation**: if TO intensity and health damage have been rising in advanced economies since the 1970s (a period coinciding with the observed productivity slowdown), the thermodynamic drag term Ω_t may account for part of the deceleration. It also offers a candidate explanation for **convergence failure**: developing economies with high τ , low H , and depleted R face effective steady states far below the standard Solow prediction, potentially constituting poverty traps. Both hypotheses are developed and tested in companion papers.

Clean growth and endogenous τ decline. It is essential to emphasize that the framework is *not* anti-growth and does not predict that τ must rise. Historical evidence shows that TO intensity has fallen substantially in many advanced economies through

energy efficiency improvements, compositional shifts toward services, regulatory pressure, and technological innovation. Between 1990 and 2020, the EU reduced greenhouse gas emissions per unit of GDP by roughly 50%. The model’s prediction is conditional: *when* τ rises (or fails to fall fast enough relative to growing output), the consequences propagate through Ω to the steady state. Clean growth—rising Y with falling τ —is not only possible within the model but is precisely the policy objective it identifies. An economy that drives τ toward zero converges to the standard Solow steady state from below, with $\Omega \rightarrow 1$. The thermodynamic framing does not imply secular pessimism; it identifies the *conditions* under which optimism is warranted.

4. Thermo-GDP

We define Thermo-GDP as:

$$\text{Thermo-GDP}_t = \text{GDP}_t - D_t = \text{GDP}_t - \lambda' \cdot \text{TO}_t \quad (21)$$

where D_t is the monetized damage from Equation (4). This is a Hicksian “green net product”: the maximum output the economy could sustain indefinitely without depleting natural and health capital stocks. It builds on and synthesizes:

- **UN SEEA-EA:** the international statistical standard for ecosystem accounting. SEEA provides the data architecture; our model provides the theoretical welfare interpretation. ^[19]
- **Adjusted Net Savings:** the World Bank’s measure of “genuine savings.” Thermo-GDP extends this by incorporating health capital. ^[21]
- **Inclusive Wealth:** the UNEP stock-based sustainability indicator. Our resource dynamics and health stock provide the flow counterparts to inclusive wealth stocks. ^[22]
- **GPI / ISEW:** earlier cost-subtraction approaches. Our framework provides microfounded shadow prices via the social-cost literature rather than ad hoc adjustments. ^[20]

The key advantage of Thermo-GDP over these predecessors is **theoretical grounding**: the damage subtraction D_t is derived from the same shadow prices that enter the modified production function, ensuring internal consistency between the growth model and the accounting framework.

5. Planetary Boundary Constraints

Rockström et al. (2009) identified nine Earth-system thresholds defining a safe operating space. ^[23] Richardson et al. (2023) report that six of nine have been

transgressed. [24,25] These can enter the model as penalty functions in the welfare objective:

$$\Phi(R_t) = \sum_n \omega_n \cdot \max(0, \bar{R}_n - R_{n,t})^2 \quad (22)$$

where ω_n weights boundary n and $(\bar{R}_n - R_{n,t})$ measures transgression. This is consistent with Raworth's "Doughnut Economics" framing [26] but provides a pathway toward formal optimization within a Ramsey-style growth framework.

6. Empirical Strategy and Testability

The framework generates several testable predictions:

- **Prediction 1:** Countries with lower TO intensity ($\tau = \text{TO}/\text{GDP}$) should exhibit higher future TFP growth, controlling for standard determinants.
- **Prediction 2:** Sector-level PM_{2.5} exposure should negatively predict labor productivity, with stronger effects in cognitively intensive industries. [12]
- **Prediction 3:** Thermo-GDP should diverge from conventional GDP most sharply in resource-intensive economies, while converging in economies undergoing green transitions.
- **Prediction 4:** Pollution reduction should generate positive growth effects exceeding the narrowly measured health cost savings (i.e., the feedback loop amplifies returns to abatement). [11,13]
- **Prediction 5:** Cross-country income convergence rates should be moderated by initial TO intensity and health capital, with high- τ , low-H countries converging more slowly.

Data sources include: SEEA accounts (27 EU member states), WHO Global Health Estimates, satellite-derived PM_{2.5} data (van Donkelaar et al.), IEA energy balances, and the World Bank's Adjusted Net Savings database. These datasets provide a feasible foundation for future cross-country calibration and simulation exercises. [19,21]

6.1 Anchor Regression: The Health–Productivity Channel

The empirically strongest entry point for testing the framework is the health–productivity feedback loop, where causal estimates and measurable elasticities already exist in the literature. We propose the following baseline panel specification, using standard econometric notation where symbols are local to this regression and distinct from model parameters elsewhere in the paper:

$$\Delta \ln(\text{TFP})_{it} = \beta_1 \tau_{it-1} + \beta_2 \ln(H_{it-1}) + \gamma' X_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

where i indexes countries, t indexes years, $\tau_{i,t-1}$ is lagged TO intensity (proxied by CO₂e emissions per unit of GDP from SEEA/IEA), $H_{i,t-1}$ is lagged health capital (proxied by HALEs from WHO, or inverse PM_{2.5} exposure from satellite data), X_{it} is a vector of standard controls (human capital, trade openness, institutional quality), α_i are country fixed effects, and δ_t are year fixed effects. The model predicts $\beta_1 < 0$ (higher waste intensity depresses future TFP) and $\beta_2 > 0$ (higher health capital boosts future TFP). Satellite-derived PM_{2.5} exposure serves as an instrument for the health channel, exploiting atmospheric transport patterns for identification. This specification can be estimated with existing data; it does not require waiting for new measurement infrastructure.

7. Theoretical Extensions of the Core Framework

The Solow model developed above is deliberately the simplest possible vehicle for the core ideas. Richer frameworks require the following extensions:

7.1 Ramsey–Cass–Koopmans

Replace the exogenous savings rate with an optimizing representative agent who chooses consumption, savings, and restoration investment. The modified Euler equation is:

$$\dot{C}/C = (1/\sigma)[f'(k) \cdot \Omega - \delta - \rho + \mu(\partial\Psi/\partial NEP)(\partial NEP/\partial C^{rest})] \quad (23)$$

where the last term captures the welfare gain from restoration investment. This extension makes the effective social discount rate *endogenous*: when environmental damage is high and restoration is productive, the model implies a lower rate than the standard Ramsey calibration, closer to Stern than Nordhaus, but derived from model structure rather than ethical assumption.

7.2 Hotelling Resource Extraction

The optimal extraction path for exhaustible resources is modified by the marginal thermodynamic damage cost. The social planner's Hamiltonian includes the shadow price of natural capital and the damage from extraction-related TO. The resulting price path differs from the standard Hotelling rule; the direction and magnitude of the deviation depend on the relative rates of damage accumulation and discount. A full Hamiltonian derivation, including the correct treatment of signs and dynamic shadow prices, is left for future work.

7.3 DSGE Environmental Shocks

TO-driven environmental shocks such as droughts, heat waves, and resource price spikes, can be incorporated as negative TFP shocks (via ϕ and ζ), labor supply shocks

(via θ), and cost-push shocks (via resource scarcity) in standard DSGE estimation toolkits. This generates a formal channel through which environmental degradation contributes to stagflation.

7.4 Optimal Carbon Price Path

The shadow prices λ in Equation (4) are equilibrium objects whose optimal time paths depend on the full model structure. Rather than a static social cost of carbon, the framework implies a dynamic price trajectory. Deriving and calibrating this trajectory is left for future work.

8. Policy Implications

8.1 Pigouvian Pricing of TO

The shadow prices λ provide the theoretically correct tax rates on each waste component. For carbon: $\sim \$190/\text{tCO}_2$.^[9] For PM_{2.5}: the labor productivity co-benefits alone may be worth $\$950\text{--}\$3,000$ per household per year per $\mu\text{g}/\text{m}^3$ reduction.^[12]

8.2 Sustainability-Weighted National Accounting

Adopt Thermo-GDP as a complementary dashboard metric alongside conventional GDP, consistent with the Stiglitz–Sen–Fitoussi Commission recommendations.^[27]

8.3 Directed Innovation Policy

Subsidize technologies that reduce τ (TO intensity). Following Acemoglu et al. (2012),^[16] early subsidies for clean technologies can permanently redirect the innovation path.

8.4 Health-Inclusive Cost–Benefit Analysis

Regulatory cost–benefit analyses should incorporate the productivity co-benefits of pollution reduction through the health capital channel. Current practice accounts for avoided mortality but typically omits the morbidity–productivity–TFP feedback loop formalized here.

9. Future Research Directions

This paper establishes the theoretical architecture. Potential companion papers include:

Optimal Carbon Price Trajectory. Derives the full time path of optimal thermodynamic prices from the social planner’s Hamiltonian. Includes the modified Hotelling rule with complete sign analysis.

Discount Rate Under Thermodynamic Constraints. Shows the effective social discount rate is endogenous to environmental conditions via the Ramsey extension.

DSGE Environmental Shocks. Embeds TO-driven shocks in the NK-DSGE framework. Tests whether post-2020 inflation contains a thermodynamic component.

Thermodynamic Drag and Secular Stagnation. Calibrates the model to US/EU/Japan 1970–2025 and tests whether rising τ correlates with the productivity slowdown.

Convergence Failure and Environmental Poverty Traps. Shows conditions under which Ω is low enough to generate poverty traps despite favorable savings rates.

10. Limitations

We acknowledge several limitations of the present framework that circumscribe the claims we make and define the work remaining for the companion papers.

First, the thermo-ecological multiplier Ω is **reduced-form**. The sustainability multiplier $\phi(\tau) = (1-\tau)\psi$ is axiomatically motivated and analytically convenient, but it is not micro-founded from optimizing firm or consumer behavior. The qualitative results are robust to alternative functional forms (any decreasing, convex ϕ with $\phi(0) = 1$ preserves the signs of all comparative statics and the amplification result), but the quantitative predictions depend on ψ , which must be estimated empirically. Until such calibrations are performed, the framework should be interpreted primarily as a qualitative and structural model rather than a quantitative forecasting engine.

Second, **TO intensity dynamics are not fully endogenized** in the present paper. We treat τ as a slow-moving state variable influenced by technology, policy, and economic structure, but we do not derive its evolution from optimizing behavior. In reality, τ responds endogenously to prices, regulation, and innovation incentives. Extensions incorporating directed technical change and Ramsey-style intertemporal optimization partially address this issue, but a fully endogenous τ pathway remains an open research problem.

Third, the **aggregation of heterogeneous waste streams** into a scalar TO intensity index is a significant simplification. Energy waste, greenhouse gases, particulate matter, and material waste have different spatial distributions, temporal profiles, and damage pathways. The weighted-norm aggregation in Equation (3) is a tractable first approximation, but it suppresses heterogeneity that may matter for policy design. Sectoral disaggregation and multi-region calibration represent important directions for future empirical work.

Fourth, **causal identification** in the health–productivity channel is challenging. While the micro-econometric literature establishes causal effects of pollution on worker productivity in specific settings, ^[11,12] extrapolating these to macro-level TFP effects

involves aggregation assumptions that have not been fully tested. The regression specification in Section 6.1 provides a starting point, but instrument validity and general equilibrium feedbacks require careful treatment.

Fifth, the model currently **abstracts from distributional effects**. Thermodynamic waste, health damage, and resource depletion fall disproportionately on lower-income populations and regions. The representative-agent Solow framework cannot capture these distributional dynamics. Heterogeneous-agent extensions would be valuable but are beyond the scope of this paper.

11. Conclusion

This paper augments the Solow growth model with three coupled state variables, including physical thermodynamic output, health capital, and natural capital, to model the economy as an open dissipative system constrained by the biosphere. The central result is the thermo-ecological multiplier Ω , which enters the steady state with amplified effect: a 10% degradation in environmental and health conditions reduces long-run output per worker by more than 10%, because it also depresses equilibrium capital accumulation.

The framework is designed to be analytically transparent. It nests the standard Solow model as a special case ($\Omega = 1$). Its predictions are testable with existing data: TO intensity predicts TFP growth, health damage feeds back to output, and dirty economies converge to lower growth paths. And its primary applied innovation, Thermo-GDP, connects directly to established accounting systems (SEEA, Adjusted Net Savings, Inclusive Wealth) while providing microfounded shadow prices.

The contribution is deliberately modest in scope: one canonical growth model, cleanly specified, with formal derivations. The questions this framework opens, including the optimal carbon price path, the endogenous discount rate, and the role of thermodynamic drag in secular stagnation and convergence failure, represent promising directions for future theoretical and empirical research.

Appendix A: Summary of Key Equations

- (1) **Production Function:** $Y = A \cdot K^\alpha \cdot L^\beta \cdot U^u$ ($\alpha+\beta+u=1$)
- (2) **Physical TO:** $TO = (E, P, M)$ — energy waste, pollution, material waste
- (3) **TO Intensity:** $\tau = \|TO\| / (p_0 \cdot Y)$ (dimensionless)
- (4) **Damage Valuation:** $D = \lambda' \cdot TO = \lambda_e E + \lambda_p P + \lambda_m M$
- (5) **Exposure Index:** $X = \kappa \cdot P / \text{Pop}$
- (6) **Health Dynamics:** $\dot{H} = \eta_0 + \eta_1 h - \delta_h H - \chi X$
- (7) **Health-Adjusted Inputs:** $L^{\text{eff}} = L \cdot \theta(H)$, $A^{\text{eff}} = A \cdot \zeta(H)$
- (8) **Net Ecological Position:** $NEP = G(R) + C^{\text{rest}} - D$
- (9) **Sustainability Multiplier:** $\phi(\tau) = (1 - \tau)\psi$
- (10) **Resource Depletion:** $dR/dt = G(R) - C$
- (11) **Modified Production:** $Y = \Omega \cdot K^\alpha (AL)^{1-\alpha}$ where $\Omega = \phi \cdot \zeta \cdot \theta^{1-\alpha}$
- (14) **Solow Dynamics:** $\dot{k} = s\Omega k^\alpha - (n+g+\delta)k$
- (16) **Steady-State Capital:** $k^* = [s\Omega^*/(n+g+\delta)]^{1/(1-\alpha)}$
- (18) **Steady-State Output:** $y^* = (\Omega^*)^{1/(1-\alpha)} \cdot [s/(n+g+\delta)]^{\alpha/(1-\alpha)}$
- (21) **Thermo-GDP:** $\text{Thermo-GDP} = \text{GDP} - \lambda' \cdot TO$
- (22) **Boundary Penalty:** $\Phi(R) = \sum \omega_n \cdot \max(0, R_n - R_n)^2$

Appendix B: Formal Derivations

B.1 Steady-State Derivation

Step 1. Begin with the fundamental Solow equation in per-effective-worker form:

$$\dot{k} = s \cdot \Omega \cdot k^a - (n + g + \delta) \cdot k$$

Step 2. At the steady state $\dot{k} = 0$, so:

$$s \cdot \Omega^* \cdot (k^*)^a = (n + g + \delta) \cdot k^*$$

Step 3. Divide both sides by k^* :

$$s \cdot \Omega^* \cdot (k^*)^{a-1} = n + g + \delta$$

Step 4. Solve for k^* :

$$(k^*)^{a-1} = (n + g + \delta) / (s \cdot \Omega^*)$$

$$k^* = [(n + g + \delta) / (s \cdot \Omega^*)]^{1/(a-1)}$$

$$k^* = [s \cdot \Omega^* / (n + g + \delta)]^{1/(1-a)} \quad \dots (16)$$

Step 5. Steady-state output per effective worker:

$$y^* = \Omega^* \cdot (k^*)^a = \Omega^* \cdot [s \cdot \Omega^* / (n + g + \delta)]^{a/(1-a)}$$

Collecting powers of Ω^* :

$$y^* = (\Omega^*)^{1 + a/(1-a)} \cdot [s / (n + g + \delta)]^{a/(1-a)}$$

$$y^* = (\Omega^*)^{1/(1-a)} \cdot [s / (n + g + \delta)]^{a/(1-a)} \quad \dots (18)$$

Interpretation: The standard Solow steady-state output is $[s/(n+g+\delta)]^{a/(1-a)}$. Our augmented model multiplies this by $(\Omega^*)^{1/(1-a)}$. Since $0 < \Omega \leq 1$ and $1/(1-a) > 1$ (for $a \in (0,1)$), the environmental penalty is amplified: if $a = 0.3$, a 10% reduction in Ω lowers y^* by approximately 14.0% ($= 1 - 0.9^{1/0.7}$).

B.2 Comparative Statics

Proposition 1. $\partial k^* / \partial \Omega > 0$.

From Equation (16): $k^* = [s\Omega^*/(n+g+\delta)]^{1/(1-a)}$. Since $1/(1-a) > 0$ and $s, (n+g+\delta) > 0$, k^* is strictly increasing in Ω^* . $\square$

Proposition 2. $\partial k^* / \partial \tau < 0$.

By the chain rule: $\partial k^* / \partial \tau = (\partial k^* / \partial \Omega)(\partial \Omega / \partial \phi)(\partial \phi / \partial \tau)$. We have $\partial k^* / \partial \Omega > 0$ (Prop 1), $\partial \Omega / \partial \phi > 0$ (since $\Omega = \phi \cdot \zeta \cdot \theta^{1-a}$ and $\zeta, \theta > 0$), and $\partial \phi / \partial \tau = -\psi(1-\tau)^{(\psi-1)} < 0$. Hence $\partial k^* / \partial \tau < 0$. $\square$

Proposition 3. $\partial k^* / \partial H > 0$.

By the chain rule: $\partial k^*/\partial H = (\partial k^*/\partial \Omega)[\phi \cdot \partial(\zeta\theta^{1-\alpha})/\partial H]$. Since $\zeta = (H/H)^m$ and $\theta^{1-\alpha} = (H/H)^{b(1-\alpha)}$, we get $\zeta\theta^{1-\alpha} = (H/H)^{m+b(1-\alpha)}$, which is strictly increasing in H . $\square$

Proposition 4 (Amplification). The elasticity of y^* with respect to Ω exceeds unity.

From (18): $\partial \ln(y^*)/\partial \ln(\Omega) = 1/(1-\alpha) > 1$ for all $\alpha \in (0,1)$. With $\alpha = 0.3$, the elasticity is 1.43. With $\alpha = 0.4$, it is 1.67. This means that environmental degradation has a leveraged effect on long-run output, because it depresses capital accumulation as well as direct production efficiency. $\square$

B.3 Transition Dynamics

The speed of convergence to steady state is determined by linearizing Equation (14) around k^* :

$$d(\ln k)/dt \approx -(1-\alpha)(n+g+\delta) \cdot [\ln(k) - \ln(k^*)]$$

The convergence rate $\lambda^c = (1-\alpha)(n+g+\delta)$ is identical to the standard Solow convergence rate. The thermo-ecological augmentation changes *where* the economy converges to (k^* is lower when $\Omega < 1$) but not *how fast* it converges. This means the growth trap is not about slow convergence, it is about converging to the wrong place.

B.4 Health Capital Steady State

Setting $H = 0$ in Equation (6):

$$0 = \eta_0 + \eta_1 h^* - \delta_h H^* - \chi X^*$$

$$H^* = (\eta_0 + \eta_1 h^* - \chi X^*) / \delta_h$$

Substituting $X^* = \kappa P^*/\text{Pop}^*$ and noting that P^* depends on Y^* and τ^* :

$$H^* = [\eta_0 + \eta_1 h^* - \chi \kappa \tau^* Y^* / \text{Pop}^*] / \delta_h$$

Since Y^* itself depends on H^* (through Ω), this defines H^* *implicitly*. The system has a fixed-point structure: H^* and Y^* must be solved simultaneously. For existence and uniqueness, we require that the feedback is not too strong: $\chi \kappa \tau$ must be small enough that the health equation's 'damage slope' does not exceed its 'recovery slope.' Under empirically plausible parameter values, this condition is satisfied and the fixed point is unique and stable.

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